

Classification Evaluation (L4 **)

Confusion matrix: rows = ground truth, cols = prediction; diagonal = correct.

Per class c (TP/FP/FN from that class):

$$P_c = \frac{TP}{TP+FP} \quad R_c = \frac{TP}{TP+FN}$$

$$F1_c = \frac{2P_c R_c}{P_c + R_c} \quad \text{Acc} = \frac{\sum_c TP_c}{N}$$

N/A cases: no predictions for $c \Rightarrow P=N/A$ (0/0); no true $c \Rightarrow R=N/A$; either undefined $\Rightarrow F1=N/A$.

Averaging:

$$\text{macro-}F1 = \frac{1}{C} \sum_c F1_c$$

$$\text{weighted-}F1 = \frac{\sum_c n_c F1_c}{\sum_c n_c} \quad (n_c = \text{support})$$

micro- $F1$: pool all TP/FP/FN then one $F1$ ($\approx \text{acc}$).

macro = each class equal (hurts on bad rare class); weighted = by frequency.

Recipe: build matrix $\rightarrow$ per-class $P, R, F1 \rightarrow$ macro=avg, weighted=avg by support. Watch 0/0.

Pitfall: accuracy can be 1.0 yet useless if test set misses classes.

Dependency Parsing + UAS/LAS (L5 **)

Arc-standard transition parser; config = (Stack, Buffer, Arcs). Stack top = B , 2nd-top = A :

- **SHIFT:** buffer front $\rightarrow$ stack.
- **LEFT-ARC(r):** arc $B \rightarrow A$ (B head of A), pop A .
- **RIGHT-ARC(r):** arc $A \rightarrow B$ (A head of B), pop B .

Repeat until buffer empty & stack=[ROOT].

$$\text{UAS} = \frac{\# \text{correct heads}}{\# \text{tokens}} \quad (\text{label ignored})$$

$$\text{LAS} = \frac{\# \text{correct head\&label}}{\# \text{tokens}}$$

Always **LAS** $\leq$ **UAS**. Right head + wrong label $\Rightarrow$ UAS yes, LAS no. (Root usually excluded.)

POS Tagging / HMM / Viterbi (L5)

Baselines: dict 85%, most-freq-tag 90%, SOTA 97%, human 98%.

HMM: states=tags(hidden), obs=words. Assumptions: transition $P(t_i|t_{i-1})$, emission $P(w_i|t_i)$. Tag via Bayes: $\hat{t} = \arg \max_t P(w|t)P(t)$.

Viterbi (DP, best path): $V_{i,t} = \max_{t'} V_{i-1,t'} P(t|t') P(w_i|t)$, cost $O(n|T|^2)$.

Encoders / BERT (L3 *)

Foundation model: self-supervised pretrain, reuse backbone.

Static (word2vec) vs **contextual** (BERT) embeddings.

$$\text{Attn}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V$$

$\sqrt{d_k}$ = numerical stability. Multi-head: A heads $d_h = d/A$. BERT-Base $d=768, A=12, d_h=64$.

FFN $W_2\sigma(W_1x+b_1)+b_2$ (768 $\rightarrow$ 3072 $\rightarrow$ 768); residual $x+\text{Sub}(x)$; LayerNorm.

BERT: encoder-only, bidirectional. MLM mask **15%**; NSP (weak, dropped later).

Fine-tune classif: prepend [CLS] $\rightarrow$ run $\rightarrow \hat{y} = \text{softmax}(Wh_{[CLS]}+b)$ $\rightarrow$ argmax.

Pooling: mean $\frac{1}{n} \sum h_i$, max.

SBERT (bi-encoder): encode sentences independently, $\cos(f(s_1), f(s_2))$, precomputable $\rightarrow$ fast. Fixes BERT $O(N^2)$ pair cost & anisotropy. Triplet loss $\max(0, d(a, p) - d(a, n) + m)$.

Cross-encoder=[CLS] s_1 [SEP] s_2 , accurate but slow (rerun each pair). Bi-encoder=fast cosine search.

RLHF (L7 **)Alignment: RLHF

Pipeline: prompts $\rightarrow$ **SFT** $\rightarrow$ human preferences $\rightarrow$ **reward model** $\rightarrow$ RL (**PPO**).

IFT/SFT: supervised on (prompt, good-resp); gives instruction-following + start policy. Self-Instruct = bootstrap data (label-first for classification).

Why beyond IFT: alignment is **comparative** (which is better?) $\rightarrow$ learn from preferences.

Reward model $r(x, y)$ scalar; can be smaller than LM; trained on pairwise comparisons $y_w \succ y_l | x$.

Bradley-Terry: $p^*(y_1 \succ y_2 | x) = \frac{e^{r(x, y_1)}}{e^{r(x, y_1)} + e^{r(x, y_2)}} = \sigma(r_1 - r_2)$, $\sigma(z) = \frac{1}{1+e^{-z}}$. Depends only on reward **difference**.

$$L_R = -\mathbb{E}_{(x, y_w, y_l)} [\log \sigma(r(x, y_w) - r(x, y_l))]$$

Describe the loss: score both responses, sigmoid of score-difference = $P(\text{winner beats loser})$; max-likelihood of observed human choices; small loss when winner scored $\gg$ loser, large when ranked wrong. Not regression/MSE: humans give relative not absolute scores.

Numbers: $r_w=2.0, r_l=1.2 \Rightarrow \sigma(0.8)=0.69$, loss $-\log 0.69=0.37$. Wrong way $\sigma(-0.5)=0.38$, loss 0.97.

PPO reward (KL-reg):

$$r(x, y) = r_\phi(x, y) - \beta D_{KL}[\pi_\theta \| \pi_{SFT}]$$

obj: $\max_\theta \mathbb{E}[r_\phi] - \beta D_{KL}[\pi_\theta \| \pi_{SFT}]$. **Role of β :** balance reward vs staying near reference. High β =stay close (safe); low β =drift, risk **reward hacking**.

KL Divergence (** compute)

$$D_{KL}(P \| Q) = \sum_x P(x) \log \frac{P(x)}{Q(x)}$$

≥ 0 ; $= 0$ iff $P=Q$; **not symmetric**. P =true/ground-truth, Q =model/measured. Use $\log_2$ for **bits** ($\log_2 z = \ln z / \ln 2$).

Ex: $P=[.5, .3, .2], Q=[.4, .4, .2]$: $.5 \log_2 \frac{.5}{.4} + .3 \log_2 \frac{.3}{.4} + 0 = .1609 - .1245 = .036$ bits.

$Q=[.85, .10, .05]: -.383+.476+.4 =$ **DPO (L8 ★)**
.493 bits (worse).

Hallucination types (L10)

Factuality = false vs the world.

Faithfulness = inconsistent with prompt/context/own reasoning.

DPO — 13-step derivation

One sentence: DPO shifts probability mass toward completions humans like and away from disliked ones, *directly* from preference pairs—no reward model, no RL.

Why: RLHF has many parts (reward model + PPO loop + instability). DPO = simple classification-style loss, stable, cheap.

Steps:

1. Start KL-reg obj: $\max_{\pi} \mathbb{E}[r] - \beta D_{KL}[\pi \| \pi_{ref}]$.
2. Expand KL $\rightarrow \mathbb{E}[r - \beta \log \frac{\pi}{\pi_{ref}}]$.
3. To min: $\min_{\pi} \mathbb{E}[\log \frac{\pi}{\pi_{ref}} - \frac{1}{\beta} r]$.
4. Define **partition fn** $Z(x) = \sum_y \pi_{ref}(y|x) e^{r/\beta}$.
5. Target $\pi^* = \frac{1}{Z(x)} \pi_{ref} e^{r/\beta}$ (valid dist).
6. Obj $\rightarrow \min_{\pi} \mathbb{E}[D_{KL}(\pi \| \pi^*) - \log Z(x)]$.
7. $\log Z$ indep of $\pi \Rightarrow \min D_{KL}(\pi \| \pi^*)$; $KL \geq 0 \Rightarrow \pi = \pi^*$.
8. Invert: $r(x, y) = \beta \log \frac{\pi^*}{\pi_{ref}} + \beta \log Z(x)$.
9. Bradley-Terry: only reward **differences** matter.
10. $Z(x)$ **cancels** in the difference (same for y_w, y_l).
11. $p(y_w \succ y_l | x) = \sigma\left(\beta \log \frac{\pi^*(y_w)}{\pi_{ref}(y_w)} - \beta \log \frac{\pi^*(y_l)}{\pi_{ref}(y_l)}\right)$.
12. Parameterize $\pi^* \rightarrow \pi_{\theta}$.
13. DPO loss (below).

$$L_{DPO} = -\mathbb{E}\left[\log \sigma\left(\beta \log \frac{\pi_{\theta}(y_w|x)}{\pi_{ref}(y_w|x)} - \beta \log \frac{\pi_{\theta}(y_l|x)}{\pi_{ref}(y_l|x)}\right)\right]$$

Closed-form optimal policy: $\pi^* = \frac{1}{Z(x)} \pi_{ref} e^{(1/\beta)r}$.

DPO — Puccinelli Q&A (memo-rize)

- **Why introduce $Z(x)$?** to **normalize** the reward-tilted π_{ref} so π^* sums to 1 (we shift mass $\Rightarrow$ must renormalize).
- **Why does it disappear?** BT uses reward **difference**;

$\beta \log Z(x)$ depends only on x (same prompt) $\Rightarrow$ identical for $y_w, y_l \Rightarrow$ cancels.

- **Why need it gone?** $Z(x) = \sum$ over **all completions** = intractable.
- **Why "normalization constant"?** rescales weights to sum 1; constant in y but **changes with prompt x** .
- **Role of β ?** strength of reward tilt vs reference proximity (implicit KL).
- **Role of win term?** $\log \frac{\pi_{\theta}(y_w)}{\pi_{ref}(y_w)}$ raises preferred response prob vs ref; loser term lowers rejected.

Benchmarking (L10 ★)

Benchmark = dataset(s)+metric.

GLUE multi-task NLU.

Task $\rightarrow$ metric: CoLA $\rightarrow$ **MCC**;
 SST-2 $\rightarrow$ acc; MRPC $\rightarrow$ acc/F1;
 STS-B $\rightarrow$ **Pearson/Spearman**;
 MNLI/QNLI/RTE/WNLI $\rightarrow$ acc.

$$MCC = \frac{TP \cdot TN - FP \cdot FN}{\sqrt{(TP+FP)(TP+FN)(TN+FP)(TN+FN)}}$$

Uses all 4 cells (F1 ignores TN). Ex: 90/10, always "yes": acc=.90 but **MCC=0**.

$\rho = 1 - \frac{6 \sum d_i^2}{n(n^2-1)}$, d_i =rank diff. Reversed ranks $\Rightarrow \rho = -1$.

NLI: premise+hyp $\rightarrow$ entail/contradict/neutral.

RTE=binary. WNLI=Winograd pronoun.

Evolution: GLUE $\rightarrow$ SuperGLUE $\rightarrow$ Hella subj $\rightarrow$ MMMU(multimodal). **Con-**

tamination: model saw test online $\rightarrow$ inflated. **LiveBench:** rolling, objective scoring.

Text-to-Speech & Tokenization 3/4

Text-to-Speech (L14)

Two stages (trained separately):

1) text→**mel-spectrogram** (Tacotron 2: char emb→conv→biLSTM that directly to decoder, 80-dim log-mel + stop).

2) mel→waveform = **vocoder** (WaveNet: AR per-sample $P(x_t|x_{<t})$, dilated causal conv, μ -law 256 cls).

TTS speaker-dependent; ASR speaker-indep. Needs text normalization.

MOS: humans rate naturalness 1–5, MOS=mean. **A/B test**: same sentence, pick better, tally. No good automatic TTS metric. mel $m = 2595 \log_{10}(1+f/700)$.

Tokenization & Embeddings (L1)

Subword balances OOV vs seq length.

BPE: merge **most frequent pair** iteratively. **WordPiece**: best score $\frac{P(xy)}{P(x)P(y)}$, “##”. **Unigram**: drop low-impact tokens. SentencePiece: raw text, space=“_”.

n-gram (m): $P(w_k|w_{k-m..k-1})$, MLE=counts; zeros→smoothing/backoff.

$PP(W) = P(w_1..w_N)^{-1/N} = 2^{H(W)}$
lower=better; ↓ as order ↑.

word2vec: CBOW (predict center, fast) vs Skip-gram (predict context, more samples). SGNS neg-sample $\propto f^{0.75}$. fastText=char n-grams.

MT / Transformers (L2)

RNN $h_t = g(Uh_{t-1} + Wx_t)$; vanish/explode→LSTM gates (forget/input/output).

Seq2seq bottleneck = final encoder state. **Attention**: $\alpha = \text{softmax}(e)$, $\text{ctx} \sum \alpha_i h_i$.

Transformer: $Z = \text{softmax}(QK^\top / \sqrt{d_k})V$; multi-head (8, $d=512$, $d_k=64$); masked decoder self-attn ($-\infty$ above diag); cross-attn (Q dec, K,V enc); **positional enc** (attn is permutation-invariant); FFN+residual+LN.

Train parallel (teacher forcing); infer autoregressive. KV-cache;

GQA/MQA share K/V.

Trap: attn for token i uses query q_i + **all keys**.

Text Generation / Decoding (L6)

Greedy argmax; **beam** keep top- k paths (direct: MT/summary).

Temperature $P \propto e^{u_i/t}$: $t < 1$ sharpen, $t > 1$ flatten.

Top-k: among k highest (fixed).

Top-p: smallest set cum $\geq p$ (adaptive). Sampling=open-ended.

LoRA: $h = x(W + AB)$, freeze W , params $r(d+k) \ll dk$. QLoRA = +4-bit base.

Instruction-tune: CE on (instr, resp), mask instr tokens.

ICL, RAG, Agents, ASR 4/4

In-Context Learning (L9)

ICL: condition with examples, **no weight update**; emergent at scale. zero/one/few-shot.

Roles: System (rules), User (task/data), Assistant (outputs).

CoT “think step by step” → better math/logic.

Self-consistency=sample $\times N$

+ majority vote.

Re-Act=Thought→Action→Observation.

Tips: simple prompts, instructions over many constraints, document (model,temp,top-p).

RAG (L11)

Retrieve chunks → augment prompt → generate. Fixes stale/parametric knowledge w/o retraining.

Pipeline: chunk→embed→index(vector DB)→embed query→similarity→prompt

Bi-encoder/SBERT retrieve (fast, precompute); **cross-encoder** rerank (accurate, slow).

$\cos(x, y) = \frac{x^\top y}{\|x\| \|y\|}$; normalized $\Rightarrow \cos \equiv \text{dot}$

rank; $\|x - y\|^2 = 2 - 2 \cos$.

Sparse: **BM25**. Hybrid: **RRF** $\sum_i \frac{1}{k + r_i(d)}$. ANN: IVF (k-means), HNSW (graph).

MRR = $\frac{1}{|Q|} \sum \frac{1}{\text{rank}_i}$; **NDCG** = $\frac{DCG}{IDCG}$, $DCG = \sum \frac{2^{rel_i} - 1}{\log_2(i+1)}$.

RAGAS: Faithfulness = (claims supported)/(claims in answer).

Agentic AI (L12)

Agent: sets goals, plans, **takes actions** (vs chatbot responds). Memory: short=context, long=vector DB/RAG. Tools: search/code/APIs.

ReAct loop. Evaluate **full transcript** (not just final).

pass@k=success over k trials.

Risks: cascading failures, over-privilege, reward hacking.

Speech-to-Text / ASR (L13)

audio→log-mel (~ 80 ch, 25ms). $dB = 10 \log_{10}(I/I_{\min})$.

wav2vec2: CNN+Transformer, self-supervised contrastive + product quantization, fine-tune **CTC**.

CTC: blank token; many

alignments→one output (collapse repeats, drop blanks); loss $-\log \sum_{align} \prod_t p(a_t|h_t)$; add LM at decode.

Whisper: enc-dec, weak supervision (680k h), multitask; robust, can loop.

$$\text{WER} = \frac{S+D+I}{N}$$

Numeric quick-reference (calc)

Sigmoid $\sigma(z) = \frac{1}{1+e^{-z}}$:
 $\sigma(0)=.50$, $\sigma(.5)=.62$, $\sigma(.8)=.69$,
 $\sigma(1)=.73$, $\sigma(1.5)=.82$, $\sigma(2)=.88$.
 $\sigma(-z)=1-\sigma(z)$.

$-\log(\cdot)$ (NLL): $-\log .69=.37$,
 $-\log .5=.69$, $-\log .38=.97$,
 $-\log .9=.105$.

log₂: $\log_2 1.25=.322$, $\log_2 .75=-.415$,
 $\log_2 1.5=.585$, $\log_2 2=1$,
 $\log_2 3=1.585$, $\log_2 4=2$. ($\log_2 z = \ln z / \ln 2$, $\ln 2=.693$.)

Worked: classification

5 sentences, classes CH/FR/DE.
 GT→Pred: CH→CH, FR→DE,
 DE→CH, CH→FR, FR→FR.

CH: TP1,FP1,FN1⇒ $P=R=\frac{1}{2}$, $F1=\frac{1}{2}$.
 FR: same $F1=\frac{1}{2}$. DE: TP0⇒
 $F1=0$.

Supports CH2,FR2,DE1. macro= $\frac{.5+.5+0}{3}=\frac{1}{3}$. weighted= $\frac{2(.5)+2(.5)+0}{5}=.4$.

Worked: UAS/LAS

6 tokens, (gold head,label | pred):
 t4 right head wrong label, t5 wrong
 head; rest correct.

Correct heads = 5/6 ⇒ **UAS**=
 .833. Correct head+label = 4/6 ⇒
LAS=.667.

BLEU (L1/L2)

$\text{BLEU} = BP \cdot \exp(\sum_n w_n \log p_n)$,
 $BP = \min(1, e^{1-r/c})$, p_n =**clipped**
 n-gram precision (cap match count
 by ref count). Precision-based;
 ROUGE=recall-based.

Worked: RLHF / MCC / Spearman

Reward NLL: $r_w=2.0, r_l=1.2$:
 $P=\sigma(0.8)=.69$, loss $-\log .69=.37$
 (good). $r_w=1.0, r_l=1.5$:
 $\sigma(-.5)=.38$, loss .97 (bad).

MCC: 90 ok/10 not, always "ok":
 TP90,FP10,TN0,FN0; acc=.90 but
 $\text{MCC} = \frac{90 \cdot 0 - 10 \cdot 0}{\dots} = 0$.

Spearman: $R = [1, 2, 3, 4, 5]$,
 $S = [5, 4, 3, 2, 1]$: $d^2 =$
 $[16, 4, 0, 4, 16], \Sigma=40$; $\rho = 1 - \frac{6 \cdot 40}{5 \cdot 24} = -1$.

Worked: retrieval / WER

NDCG@5, $\text{rel} = [0, 3, 2, 1, 0]$:
 $\text{IDCG} = 0 + \frac{2^3-1}{\log_2 3} + \frac{2^2-1}{\log_2 4} + \frac{1}{\log_2 5} + 0 =$
 $\frac{1}{2} + 4.42 + 1.5 + 0.43 = 6.35$;
 $\text{IDCG} = 9.39 \Rightarrow \text{NDCG} = .68$.

WER: ref 10 words, 1S+1D+2I
 $\Rightarrow (1+1+2)/10 = .40$.

Cosine: unit vecs, $\cos=.6 \Rightarrow$
 $\|x-y\|^2=2-1.2=.8$.

LoRA: $4096^2=16.7\text{M}$ full; $r=4 \Rightarrow$
 $4(4096+4096)=32768$ ($\sim 512 \times$
 fewer).

Fact bank / traps

- Lower perplexity = better LM (↓ with n-gram order).
- BPE merges **frequent pair**; WordPiece by **score**; Unigram **prunes**.
- Temp $t<1$ sharpen, $t>1$ random; top- k fixed, top- p adaptive.
- Attn weights: **query q_i vs all keys**; $\sqrt{d_k}$ =stability.
- BLEU=precision+BP; ROUGE=recall.
- KL: not symmetric, ≥ 0 , = 0 iff equal; use **bits**.
- LAS≤UAS always.
- Accuracy misleads on imbalance / missing classes (use macro-F1, MCC).
- DPO: no reward model, no RL; partition fn cancels.
- Bi-encoder=retrieve(fast); cross-encoder=rerank(accurate,slow).
- RLHF reward = $r_\phi - \beta \text{KL}$; high β =stay near SFT.

Exam tactics

Need only **3.5/6** (5.5 pregrade @25%). Bank MC + quick calcs first, then text answers. State assumptions if unclear. KL in **bits** ($\log_2$). Show formula+plug-in. **Must-compute**: P/R/F1, macro/weighted, UAS/LAS, KL(bits), reward/DPO sigmoid-NLL, MCC, Spearman, WER, perplexity, cosine, NDCG.